
# **W1-03:** Service Needs

COMPSCI4064 (H) and COMPSCI5088 (M)

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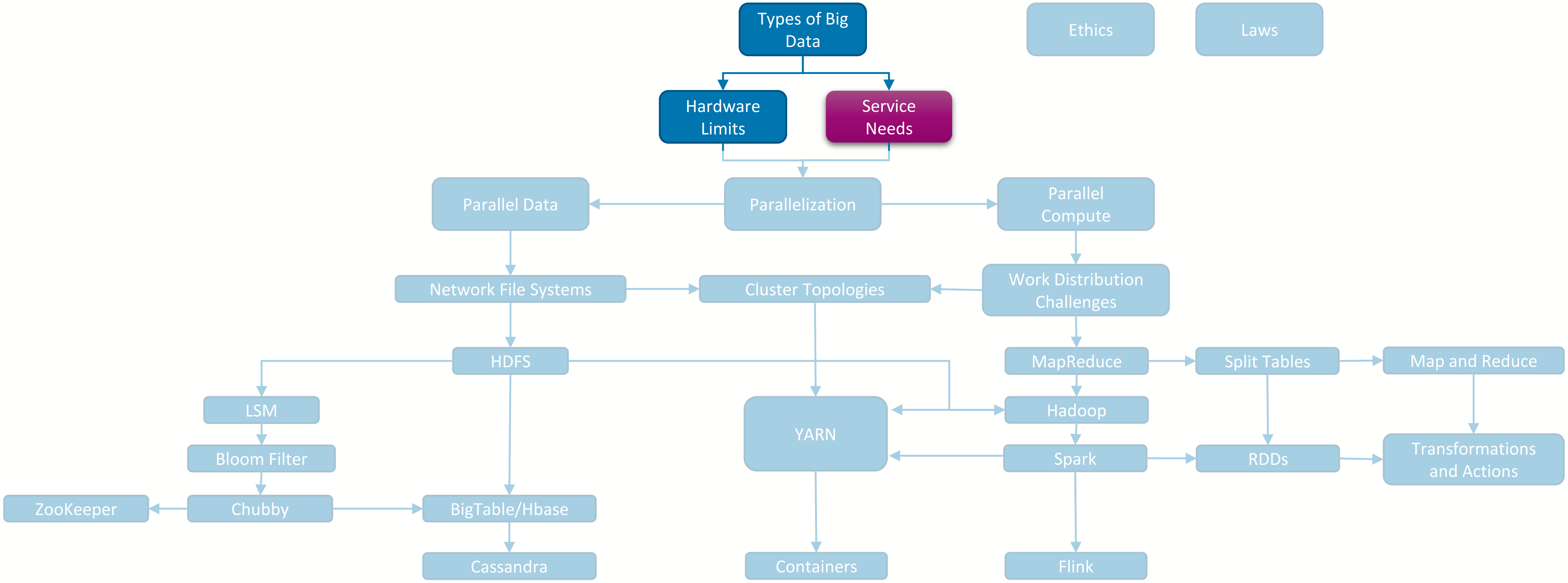
# Where are we?



Types of Big Data
Ethics
Laws
Hardware Limits
Service Needs
Parallel Data
Parallelization
Parallel Compute
Network File Systems
Cluster Topologies
Work Distribution Challenges
HDFS
MapReduce
Split Tables
Map and Reduce
LSM
YARN
Hadoop
Bloom Filter
Spark
RDDs
Transformations and Actions
ZooKeeper
Chubby
BigTable/Hbase
Cassandra
Containers
Flink

# What we use Big Data for

- Big Data is defined by its *Use*
  - “**capturing, storage, curation, management**” – how we store and access it
  - “**processing**” – how we can use it to gain value

- At a high level, there are three main Big Data use-cases
  - **Extraction/Filtering**: I have a large silo of data and I need to extract a subset of it matching some criteria
  - **Transformation**: I have a function that I want to apply to each item in my data and save the outcome
  - **Aggregation**: I want to merge (and typically concurrently transform) a set of data items

# Batch vs. Streaming

- Another important dimension to consider is whether we are working with largely static data, or new data is arriving over time

- The underlying Big Data solution you use will be different in these two cases
  - **Static Data**: Tackled with Batch processing, where large blocks of that data are analysed at once, usually with multiple blocks being processed concurrently, that are subsequently merged together
    - Cares about completion time
  - **Stream Data**: Tackled with Stream processing, where items are processed individually (or in very small blocks) within a pipeline of stages, where multiple items can be processed concurrently within the pipeline
    - Cares about processing latency per item

# Examples

| | Extraction/Filtering | Transformation | Aggregation |
|---|---|---|---|
| Batch | • SQL Select<br>• Batch Search | • Search Indexing<br>• Sentiment Analysis<br>• Machine Learning | • SQL JOIN<br>• Word counting<br>• Clustering |
| Stream | • Stream Filtering | • Transaction Fraud Detection<br>• User Interaction Logging<br>• Live Recommendation | • Time Window Analytics<br>• Stream JOIN |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